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Influence of soil nutrients on plant characteristics and soil hydrological responses

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The effects of nutrients on plant growth have been widely studied. In agricultural research, some studies have showed significant root growth in high-nutrient-supplied soil, however, some other studies, reported better lateral root growth in low-nutrient-supplied conditions. However, the effects of NPK (nitrogen, phosphorous, potassium) water-soluble fertilisers on plant roots and shoot growth in bioengineered soil slopes and their influence on soil suction and water-retention ability are still unclear. This paper quantifies the growth of a tree in NPK nutrient-supplied soil and its effects on tree-induced soil suction and water-retention ability during evapotranspiration. Three replicates of *Schefflera heptaphylla* (Ivy tree) were grown for 6 months in nutrient-poor and nutrient-supplied compacted soils typical of bioengineered slopes and embankments. Plant characteristics, induced suction and volumetric water content (VWC) were monitored throughout the growth period. After 6 months of growth, leaf area index (LAI) and peak root area index (RAI) increased by 350 and 133%, respectively, in nutrient-supplied soil compared with the nutrient-poor soil. This is because nitrogen stimulated chlorophyll synthesis enabling plants to produce larger leaves. Additionally, nitrogen mediated phosphorous to be utilised by roots in soil to induce growth of fine roots and thus increase the root surface area and root volume in soil pores. After 3 days of drying, the vegetated nutrient-supplied soil induced 15–50 kPa higher suction than the vegetated nutrient-poor soil due to the higher RAI, LAI and root volume of the plants grown in nutrient-supplied soil, which enable plants to absorb and transpire more water. In contrast, the water-retention ability reduced in the nutrient-rich vegetated soil because more clustered fine roots in soil pores decreased the pore diameter and increased suction.

KEYWORDS: soil stabilisation; suction; vegetation

NOTATION

C	control
D_{10}	grain diameter at 10% passing
D_{30}	grain diameter at 30% passing
D_{60}	grain diameter at 60% passing
N	nutrient
NPK	nitrogen, phosphorous and potassium
m	fitting parameter in van Genuchten's equation (van Genuchten, 1980)
n	fitting parameter in van Genuchten's equation (van Genuchten, 1980)
α	fitting parameter in van Genuchten's equation (van Genuchten, 1980)
θ_r	residual VWC
θ_s	saturated VWC

INTRODUCTION

Vegetation has been widely used as a sustainable and environmentally friendly engineering practice for stabilising slopes and landfill covers through hydrological reinforcement (Cazzuffi *et al*., 2006; Sinnathamby *et al*., 2013). Transpiration by plants increases soil suction, with a consequent reduction in soil permeability and increase in soil shear strength (Ng & Menzies, 2007; Ng & Leung, 2012). Root traits can both influence soil shear strength (Mickovski *et al*., 2009) and induce substantial change in soil water-retention characteristics (Leung *et al*., 2015). However, soil characteristics (i.e. physical and chemical composition) affect vegetation growth and hence the effectiveness of roots on slopes and landfill covers. Roots will be denser by proliferation in nutrient-rich soil (Stokes *et al*., 2009). Soil with high-nutrient supply usually provides healthier species in terms of growth (Arredondo & Johnson, 1999) and root traits reflect an increased plant development (Gersani & Sachs, 1992). It is well known that fine roots have high surface area and thus are mostly responsible for water and nutrient uptake from soil (Eamus *et al*., 2016). In contrast, some studies showed significant lateral root elongation in low nitrate or phosphorous supplied soil (Johnson *et al*., 1996; Zhang & Forde, 1998). NPK granular fertilisers were used for bioengineering techniques (vegetated man-made slopes) (GEO, 2011), which had long-term significant effects on growth of *Schefflera heptaphylla* on the degraded hillside grassland in Hong Kong (Hau & Corlett, 2003). This plant is a recommended native species for erosion control planting in Hong Kong (GEO, 2011), which can survive in man-made slopes (Garg *et al*., 2015a, 2015b, 2015c). However, there is a lack of contribution whether nitrogen-rich NPK water-soluble fertilisers can stimulate both shoot and root growth of plants in heavily compacted soil, which is common for man-made slopes in countries such as the United States (TDOT, 1981) and Hong Kong (GCO, 2000). Moreover, how nutrients will affect the root proliferation in heavily compacted soil and how the proliferated denser roots by nutrient supply will affect the induced soil suction and water-retention ability is not well understood.

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This paper quantified the effects of NPK water-soluble fertilisers on the growth of *S. heptaphylla* (Ivy tree) in heavily compacted (RC-95%) silty sand and its effects on induced soil suction and water-retention ability under controlled and known atmospheric parameters rather than the field environment (variable parameters) to isolate the effects of nutrients on plants.

TEST PLAN, SET-UP AND INSTRUMENTATION

Two test series were conducted to grow plants in heavily compacted soil with and without nutrient supply and to investigate soil suction and volumetric water content (VWC). The first series was used as reference without using any nutrient in soil and the second series was conducted by supplying NPK (30-10-10) water-soluble nutrient in soil.

Figure 1 shows the schematic set-up of a vegetated soil column. Six columns were constructed with an inner diameter of 200 mm and a height of 400 mm. Completely decomposed granite (CDG) soil was compacted up to 390 mm depth at a relative compaction (RC) of 95% (corresponding to the dry density of 1777 kg/m^3 and 12% moisture content). All measured index properties and concentrations of nitrogen, phosphorous and potassium of the CDG soil are summarised in Tables 1 and 2, respectively. Individual *S. heptaphylla* was transplanted at the centre in six columns with similar basal diameter (10 ± 2 mm) and root depths (125 ± 10 mm) for fair comparison. Four miniature tip tensiometers at 50, 130, 210 and 290 mm depth (from soil surface) were installed at the centre of the columns. SM-300 probes were installed at 50 and 130 mm depth right next to the tensiometer to monitor the VWC. The purpose was also to investigate the effects of soil nutrition on root growth and occupancy in soil pores with consequent soil aggregation and changes in soil-pore structure, which might affect soil water-retention behaviour. All columns were placed under temperature (25 ± 1°C) and humidity (55 ± 5%) controlled room under a cool white fluorescent lamp of 120 (μmol/m^2)/s intensity for the whole testing period.

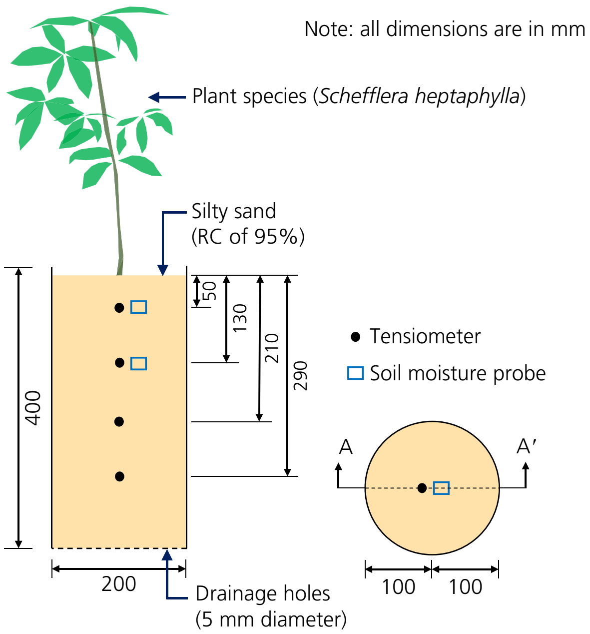

Fig. 1. Typical schematic set-up and instrumentation of a tree-vegetated column in (a) cross-section view A–A′ and (b) plan view

Table 1. Index properties of CDG

Index properties	Value
Standard compaction tests	
Maximum dry density: kg/m^3	1870
Optimum moisture content: %	12
Particle-size distribution	
Gravel content (>2 mm): %	19
Sand content (≤2 mm): %	42
Silt content (≤63 μm): %	27
Clay content (≤2 μm): %	12
D_{10}: mm	0·15
D_{30}: mm	0·7
D_{60}: mm	2
Coefficient of uniformity (D_{60}/D_{10})	13·3
Coefficient of curvature ($(D_{30})^2/(D_{60}D_{10})$)	1·6
Specific gravity	2·60
Atterberg limit	
Plastic limit: %	26
Liquid limit: %	44
Plasticity index: %	18
Unified Soil Classification System (USCS)*	Silty sand (SM)

*ASTM (2010).

Table 2. Atomic and mass concentration (%) of major nutrient elements (N, P, K) in CDG soil

Major nutrient elements	Atomic concentration: %	Mass concentration: %
Nitrogen (N)	0·19 ± 0·04	0·13 ± 0·03
Phosphorous (P)	0	0
Potassium (K)	0·48 ± 0·07	0·94 ± 0·13

TEST PROCEDURE

For each test series, three replicates of plants (total six plants) were grown for 6 months and the suction and VWC were monitored. Every 2 days, all columns were irrigated with the same amount of water and every 8 days, three columns were instead irrigated with NPK (30-10-10) nutrient mixed with water. NPK (30-10-10) is a water-soluble fertiliser and was used in three vegetated soil columns after mixing 2 g of NPK (30-10-10) fertiliser with 1 litre of water. The cumulative fertiliser used for each column over 6 months was 15 g (corresponding to 212 kg/ha). The decreasing suction values at 50 and 130 mm depth (monitored during irrigation by tensiometers) ensured that the nutrient solution reached up to the roots. At the end of 3 and 6 months, in both series, ponding head was applied on the surface of the bare and six vegetated columns until suctions at all four depths were 0 kPa and percolation through the holes at the column base was observed. Then the columns were exposed to the environment to quantify the effects of evapotranspiration by plant–soil system on suction responses. During the growth period, leaf area index (LAI) was monitored and after testing, plants were carefully removed from the columns and RAI was determined using image J as per the procedures described in Garg *et al.* (2015a, 2015b, 2015c). Stem, leaves and roots were weighed and oven-dried at 60°C for 24 h and again weighed to determine the dry biomass (Liang *et al.*, 1989). Finally, the root–shoot biomass ratio was calculated for each plant. The ratio of the root biomass and soil volume and the ratio of the root volume and soil volume for all vegetated nutrient-poor and nutrient-supplied soil columns are shown in Table 3.

INTERPRETATION OF TEST RESULTS

Figure 2 compares the LAI between 'C' and 'N' tests. In the N test, LAI increased by 350% compared to the C test after 6 months of plant growth. Tingey *et al.* (1996) also found 40% increase of LAI using nitrogen fertilisers (100 kg N/ha) in clay–loam soil after 6 months of plant growth from ponderosa-pine seeds. Nitrogen insufficiencies reduce LAI resulting in reduced surface light interception for photosynthesis (Cechin & Fumis, 2004). CDG soil did not have sufficient nitrogen (Table 2) and is generally considered as nutrient-poor soil (SILTech, 1991). However, the growth rate of leaves is larger when nitrogen-rich nutrient is present in soil because nitrogen plays a significant role in chlorophyll synthesis, which stimulates plant growth resulting in larger leaves (Jasso-Chaverria *et al.*, 2005).

Figures 3(a) and 3(b) compare the root geometry of plants growing for 6 months in nutrient-poor and nutrient-supplied soil, respectively. Without nutrient supply, the roots are not as dense as the roots growing in nutrient-supplied soil. The roots (Fig. 3(b)) are denser with clustered fine roots because in nutrient-rich soil-patch, roots proliferate with more second- and third-order lateral roots (Stokes *et al.*, 2009).

Figure 4 compares RAI distributions along the root depth of plants growing in nutrient-poor and nutrient-supplied soil after 6 months of growth. RAI distributions are non-linear and parabolic in shape which is consistent with the research study by Garg *et al.* (2015a, 2015b, 2015c) and Ng *et al.* (2016), where the same plant species, soil type and soil density were used. The root surface area and hence, RAI was larger at 70–80% depths of root length, also observed by Garg *et al.* (2015a, 2015b, 2015c) during the root distribution factor (R_{df}) measurements. The RAI difference between C and N tests was significant up to 130 mm depth

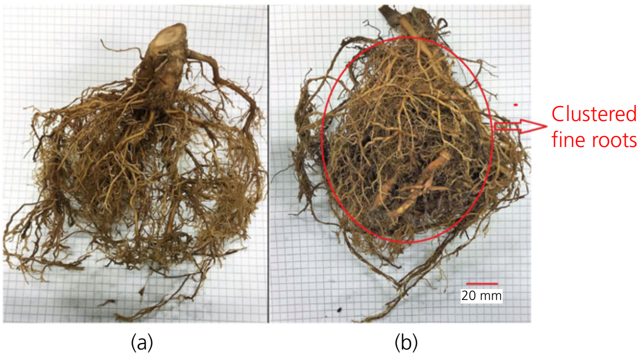

Fig. 3. Root growth of plants after 6 months in (a) nutrient-poor and (b) nutrient-supplied soil (the sheet of paper in the background contains 5 mm × 5 mm squares)

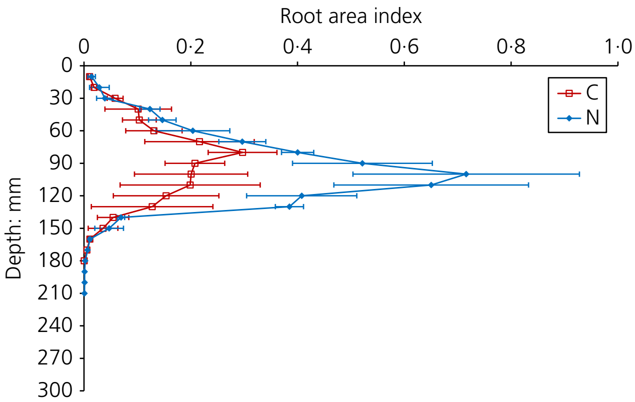

Fig. 4. Measured changes in RAI of plants after 6 months of growth in nutrient-poor and nutrient-supplied soil (C represents the controlled test without nutrient supply and N represents the test with nutrient supply)

Table 3. Normalised root biomass and root volume of the vegetated soil columns

Test	Root biomass: g/soil volume: m^3	Root volume: m^3/soil volume: m^3
C	775·5 ± 195·9	0·04 ± 0·02
N	1583·7 ± 800	0·088 ± 0·04

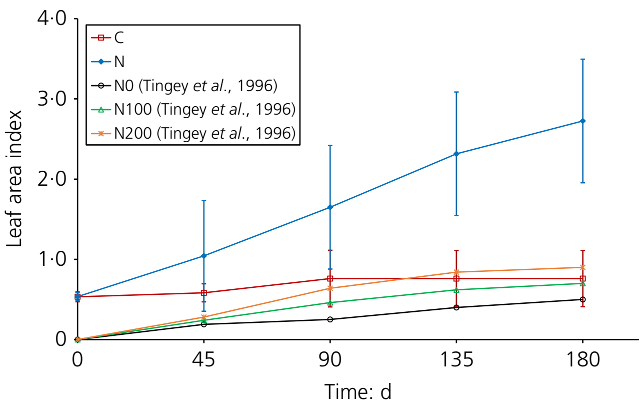

Fig. 2. Measured changes in LAI of plants before transplantation and during 6 months of growth in nutrient-poor soil and nutrient-supplied soil (C represents the controlled test without nutrient supply and N represents the test with nutrient supply)

(desired nutrient-rich zone) since the initial root depth was within 140 mm. The peak RAI difference was about 130% implying that the total root surface area and the root volume was significantly larger over a certain cross-section area and in a certain volume of soil in nutrient-supplied vegetated soil columns (Table 3). This is because phosphorous (P) enhances root growth, particularly lateral fine root development (Brady & Weil, 2002), which were clustered as indicated in Fig. 3(b).

In contrast, several research studies have shown contrasting responses of lateral root elongation due to soil nitrate and phosphorous concentration (Johnson *et al.*, 1996; Zhang & Forde, 1998). Since different plants and their genes respond to nutrient-rich soil regions in different ways, this study implies that nitrogen and phosphorous supply has significant effects on the plant roots of *S. heptaphylla* to stabilise bioengineered slopes.

Figure 5 shows the drying path of the soil water-retention curve (SWRC) of bare, nutrient-poor and nutrient-supplied vegetated soil obtained by relating VWC to suction at a depth of 50 mm during drying from three replicates. Data points were scattered in a wider range in nutrient-supplied vegetated soil compared with the nutrient-poor vegetated soil due to the larger variation in the root biomass as shown in Table 3 (Leung *et al.*, 2015). van Genuchten (1980) equation was used to fit the SWRCs and the required fitting parameters are shown in Table 4. For any given suction, the rooted soils in tests C and N had noticeably greater water-retention ability than the bare soil, also observed by

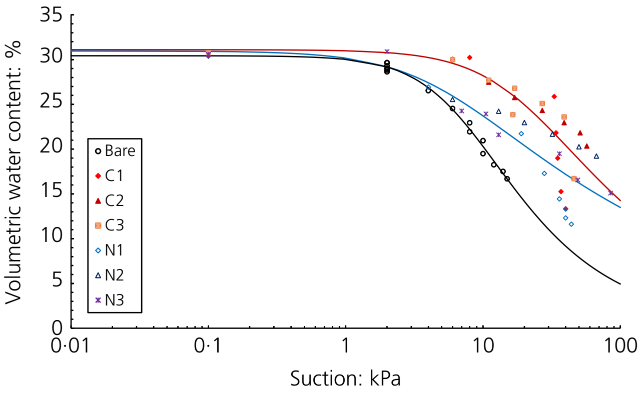

Fig. 5. Measured SWRCs of bare and vegetated soil along the drying path in nutrient-poor and nutrient-supplied soil (C1, C2 and C3 represent the controlled test of three replicates without nutrient supply and N1, N2 and N3 represent the test of three replicates with nutrient supply)

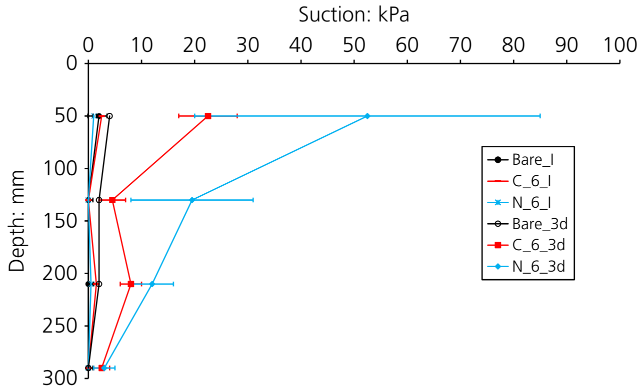

Fig. 6. Measured suction profiles of bare and nutrient-poor and nutrient-supplied vegetated soils before and after 3 days drying after the sixth month of plant growth (C represents the controlled test without nutrient supply and N represents the test with nutrient supply; 6 represents the drying test conducted at the sixth month of plant growth; I and 3d represent the initial suction in soil before drying and suction after 3 days of drying)

Table 4. Summary of fitting parameters for van Genuchten (1980) equation

Test	Drying SWRC				
	θ_s: m^3/m^3	θ_r: m^3/m^3	α: m^{-1}	n	m
Bare	0·31	0·00	13·60	1·70	0·41
C	0·31	0·00	4·98	1·47	0·32
N	0·31	0·00	20·14	1·28	0·22

Leung *et al.* (2015), because roots can induce substantial changes in SWRCs. However, there was 5–6 kPa differences in the air-entry value (AEV) and water-retention ability between C and N tests due to 50% RAI difference between the roots (Fig. 4) and more than 2 times difference between the normalised root volume (Table 3). For any given suction, the nutrient-supplied rooted soils had reduced water-retention ability compared with the nutrient-poor rooted soil because plants grown with nutrient supply has more cluster of fine roots (Fig. 3(b)), larger root surface area and hence RAI (Fig. 4) and larger root volume. This occupied and clogged the soil pores changing the soil-pore structure with consequent soil aggregation. Fine roots covered by root hairs significantly increase absorptive surface area and improve contact between roots and soil, which increases occupancy in soil pores and they are the most permeable portion of a root system having the greatest ability to absorb water (McElrone *et al.*, 2013). For a given soil water content, root occupancy in soil-pore space (4% in C test and 8% in N test (Table 3)) could reduce the diameter of the soil-pore throat changing the pore structure and, which in turn, increases suction according to the capillary law (Scanlan & Hinz, 2010). Since SWRC primarily depends on soil-pore structure and its distribution (Romero *et al.*, 1999; Ng & Leung, 2012), soil water-retention ability was different in nutrient-poor and nutrient-supplied vegetated soil. The drying curve is significant to geotechnical engineers in understanding crack formations in bioengineered slopes and landfill cover for vegetative layers that was emphasised and discussed by Gadi *et al.* (2016).

Figure 6 shows the suction profiles along the depth before and after 3 days of drying after 6 months of plant growth. The initial suctions in bare and vegetated soils were 0–2 kPa. The higher suction induced at shallow depth (50 mm) in bare and vegetated soil than deeper depths are due to the hydraulic gradient established at the soil–atmosphere interface during surface evaporation. During drying, evapotranspiration-induced suction increased by 26–50 kPa (at 50 mm depth) and by 15–26 kPa (at 130 mm depth) in nutrient-supplied vegetated soil compared with the nutrient-poor vegetated soil. This is due to 50% (at 50 mm depth) and 138% (at 130 mm depth) larger RAI (Fig. 4) and 2 times larger root volume (Table 3) besides 160–200% larger LAI in nutrient-supplied vegetated soil (Fig. 2). However, there is a variability of suction measurements in the replicates of vegetated soils which represents the uncertainty of suction behaviour during evapotranspiration due to water uptake by roots. In fact, recent studies used probabilistic approach to analyse the stability of vegetated slopes by treating suction as a random variable (Das *et al.*, 2018). Since a cluster of fine roots (Fig. 3(b)), larger root surface area (Fig. 4) and larger root volume in soil (Table 3) can absorb and uptake more water and nutrient (McElrone *et al.*, 2013; Eamus *et al.*, 2016) and higher leaf surface area (Fig. 2) can transpire more water (Kelliher *et al.*, 1995), induced suction in nutrient-supplied vegetated soil was significantly higher. This implies that, using additional NPK nutrient can stimulate leaf and root growth significantly having long-term establishment in slopes and thus higher suction can be induced in vegetated heavily compacted soil. Therefore, there would be significant impact on the performance of the infrastructure because soil shear strength will increase, soil permeability during rainfall will decrease (Ng & Menzies, 2007; Ng & Leung, 2012) and the probability of failure (P_f) will be less in the nutrient-supplied treed slopes due to higher amount of evapotranspiration-induced soil suction (Das *et al.*, 2018).

Figure 7 shows the correlation of the root–shoot biomass ratio with induced peak soil suction in nutrient-poor and nutrient-supplied vegetated soils. In both tests, higher root–shoot biomass ratio induces higher suction and correlation between root–shoot biomass ratio and induced peak suction is stronger ($R^2 = 0{\cdot}86$) than the study by Boldrin *et al.* (2017) ($R^2 = 0{\cdot}65$) because it considered ten different kinds of plant species. Steeper correlation was observed in the N test implying that the root–shoot biomass ratio and their induced suction both were significantly higher in nutrient-supplied vegetated soil. This is because increase in the root–shoot ratio allows the plant to extract more water from the soil (Stulen & den Hertog, 1993) and therefore, induced suction would be higher which can

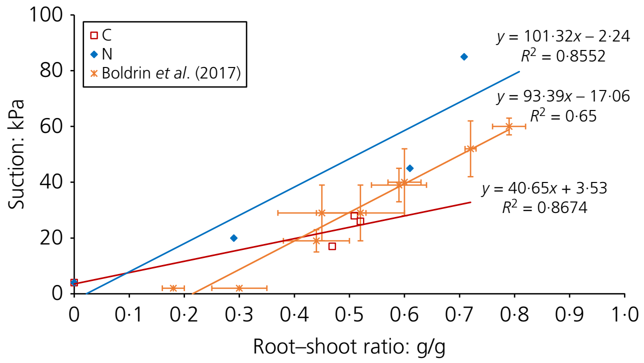

Fig. 7. Relationship of root–shoot biomass ratio with matric suction (kPa) after 3 days of drying in nutrient-poor and nutrient-supplied vegetated soil after 6 months of plant growth (C represents the controlled test without nutrient supply and N represents the test with nutrient supply; Boldrin *et al*. (2017) used ten different woody plant species in sandy loam soil)

increase hydrologic reinforcement and reduce the probability of failure (P_f) in bioengineered treed slopes (Das *et al*., 2018). This highlights the importance of considering the combined effects of both the below- and above-ground organs on the hydrologic reinforcement to soil (Boldrin *et al*., 2017).

CONCLUSIONS

This study explores the effects of NPK (30-10-10) water-soluble fertiliser on *S. heptaphylla* and its induced soil suction and VWC. Two test series (three replicates for each test) with and without NPK water-soluble fertiliser supply in heavily compacted (95% RC) silty sand were conducted for 6 months. After 6 months, LAI and peak RAI increased by 350 and 133%, respectively, and clustered fine roots and more than 2 times larger root volume were observed in nutrient-supplied vegetated soil compared with nutrient-poor soil. This is because nitrogen stimulates chlorophyll synthesis providing plants with larger leaves and mediates phosphorous to be utilised by roots in soil, which enable plants to grow more fine roots and thus larger root surface area. Due to occupancy in soil pores with more clustered fine roots, larger root surface area and root volume in a certain volume of soil which decreases pore diameter with consequent soil aggregation and increases suction by capillary law, water-retention ability decreased in nutrient-supplied vegetated soil. Because of having larger LAI, RAI, root volume and root–shoot biomass ratio, plants in nutrient-supplied vegetated soil could extract, absorb and transpire more water and so 15–50 kPa (68–173%) higher soil suction was induced after 3 days of evapotranspiration. This paper suggests that plants can survive in heavily compacted soil, but their growth is hindered if nutrient is not supplied in the soil. Additional nitrogen-rich NPK water-soluble fertiliser supply during irrigation of vegetated soil could stimulate plant growth, which increases plant-induced soil suction that can increase shear strength, reduce permeability effectively during rainfall and reduce the probability of failure of bioengineered treed slopes.

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